Actual walk characters and phase contraction

Result. Lean derives the Fourier coefficient law for the actual independent additive-walk PMF on every finite abelian group. Character expectation at time t equals $\lambda^t \cdot \psi(\text{start})$, and actual squared modulus equals $\text{normSq}(\lambda)^t$, with λ the actual weighted character sum. Complex expectation and its bind law come from normalized finite PMF probabilities and checked real expectations; no supplied spectral formula is assumed. [pcc:unverified]

Actual iid law	Character expectation = $\lambda^t \cdot \psi(\text{start})$ Actual squared character moment = $\text{normSq}(\lambda)^t$
Derived contraction	Normalized unit-phase variance = pair-defect sum / 2 Positive pair + phase separation ⇒ strict contraction
Domain controls	Prime + positive pair + distinct supports + $j \neq 0$: $\text{normSq}(\lambda) < 1$ Zero weight or composite0/2: coefficient1; j0 is trivial

Quantitative pair estimate. For actual PMF unit phases, $1 - \text{normSq}(E[z])$ equals one half of the weighted double sum of $\text{normSq}(z_i - z_j)$. Nonnegative defects and the two ordered terms for distinct labels u,v yield $\text{normSq}(\lambda) \leq 1 - 2 \cdot q_u \cdot q_v \cdot (1 - \text{Re}\ \psi(a_u - a_v))$. Positive weights then make $\text{normSq}(\lambda) > 1 - 2 \cdot q_u \cdot q_v \cdot \delta$ imply $\text{Re}\ \psi(a_u - a_v) > 1 - \delta$. Thus $\delta = 0.01$ supplies the source's0.99 phase cutoff for the chosen pair. [pcc:unverified]

Actual prime-field character. The frequency character is Mathlib's standard complex additive character at $x \cdot j$. Lean proves its literal cosine formula using the natural residue representative and proves injectivity for prime p/nonzero j. For any injective support and any two distinct positive-weight labels, every nonzero mode has $\text{normSq}(\lambda) < 1$. Character unit values and phase differences are derived, not numerical assumptions. [pcc:unverified]

Controls. Index j=0 has actual coefficient1 and every-time squared moment1. Over every prime field, the actual pure-label PMF has weights1/0 and distinct supports0/1; its nonzero-index1 coefficient is1, directly refuting strict contraction with positivity deleted. On ZMod4, actual fair positive weights and distinct supports0/2 have coefficient1 at nonzero index2, checking deletion of primality. These preserve the qualified source. [pcc:unverified]

Sources and remaining work. Hildebrand1994: original13-page read/inspection credited, Lemma3/5 on195–198 rechecked. Author preprint math/0401189v1: selected section4.2/Proposition2/explicit $j \neq 0$ on13–16, reference14p36; visual15–16 only. Official finite-character/complex APIs and actual weighted-kernel proof read. Full source minimum cutoff, Parseval/TV and averaged spectral sum remain open; corrected box asymptotic is proved separately. [pcc:unverified]

Checks. Whole gate: 37 declarations, zero failures, zero trusted native exceptions. All 79 build/processing/provenance files retained losslessly in 32 objects, five gate files and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified; KB/Conductor fallback recorded. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]